Senim Risk
AN INTERPRETABLE CREDIT-RISK SYSTEM

REPRODUCIBLE RESEARCH AND ENGINEERING STUDY

Senim Risk: An Interpretable Decision-Support Prototype for Thin-File Credit Risk

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The strongest result is not the headline score. It is the failure revealed when one repayment-history proxy is removed.

0.830	0.492	0.775
full synthetic RF ROC-AUC	RF without late-payment proxy ROC-AUC	public UCI RF ROC-AUC

Evidence boundary
Synthetic Pavlodar-oriented data support engineering and diagnostic experiments. The public UCI benchmark tests pipeline portability. Neither validates real Kazakhstan lending decisions.

Repository	Live prototype
https://github.com/alex-tereshkovv/micro-score	https://alex-tereshkovv.github.io/micro-score/

Human-in-the-loop prototype. Synthetic public demo. Not validated for real lending.

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This paper separates synthetic evidence, public benchmark evidence, product behavior, and blocked real-world claims.

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Abstract

Thin-file borrowers have too little conventional credit history for many standard underwriting processes, but alternative behavioral data can introduce new privacy, proxy, and reliability risks. Senim Risk studies this tension through a reproducible machine-learning pipeline and a working decision-support prototype. Experiment A uses 5,000 synthetic borrower records to compare Logistic Regression and Random Forest, audit leakage and proxy dependence, test feature-group ablations, examine calibration and classification errors, and compare three-zone lending policies. Repeated-split evaluation, stratified bootstrap intervals, and deterministic covariate-shift stresses test whether the central ablation result is stable. Experiment B applies the same evaluation discipline to the public UCI Default of Credit Card Clients dataset. On the synthetic held-out set, Random Forest reaches ROC-AUC 0.830 and Logistic Regression reaches 0.806. However, `late_payment_count` alone reaches directional ROC-AUC 0.827, and removing it reduces both models to approximately random ranking (0.486-0.492). On the UCI benchmark, Random Forest reaches ROC-AUC 0.775 and Brier score 0.159. These results show that a credible score cannot be judged from one headline metric: feature provenance, ablation, calibration, policy effects, and deployment context are equally important. Senim Risk is therefore presented as a human-in-the-loop research prototype, not a validated lending model.

Keywords: alternative credit scoring; thin-file borrowers; interpretable machine learning; ablation; calibration; decision thresholds; responsible AI

1. Introduction

Credit reporting can exclude people and small businesses that lack documented income, bank accounts, or a sufficiently detailed repayment record. Alternative data may help construct a fuller view of thin-file applicants, but it can also amplify inaccurate, opaque, or discriminatory signals [1, 2]. A technically strong classifier is therefore not sufficient. A lending support system must also reveal what drives the score, how sensitive the result is to individual features, what policy converts a probability into an action, and where the evidence stops.

Senim Risk investigates these questions in the context of a possible regional microfinance workflow in Pavlodar, Kazakhstan. The project combines two layers:

- a research pipeline for modeling, ablation, calibration, error analysis,

segment review, and public benchmark validation;

- a product prototype with borrower, analyst, and administrator workflows,

model provenance, audit events, threshold policies, and portfolio simulation.

The intended contribution is not a new state-of-the-art scoring algorithm. It is a transparent engineering study of how an apparently promising model can be challenged before it is trusted.

2. Research Questions

The study asks four questions:

1. How well do interpretable linear and nonlinear baseline models rank synthetic

credit risk after obvious leakage-like fields are removed?

2. Does the result survive removal of repayment-history and feature groups that

may not exist for genuinely thin-file borrowers?

3. How do probability thresholds change approval access, manual-review load,

and exposure to high-risk cases?

4. Does the main ablation result survive repeated splits, held-out resampling,

and controlled input-availability stresses?

5. Can public data test portability without proving local validity?

3. Background And Related Work

The World Bank and International Committee on Credit Reporting describe alternative data as a possible route to richer credit profiles for thin-file or credit-invisible customers, while emphasizing consent, data quality, discrimination, cybersecurity, and explainability risks [1, 2]. This motivates Senim Risk's decision to treat alternative data as evidence to be audited, not as automatically fairer data.

The public benchmark originates from the UCI Default of Credit Card Clients dataset and the accompanying study by Yeh and Lien [3, 4]. It contains real Taiwan credit-card records and supports comparison with a recognized binary default-prediction task. Its geography, borrower type, and product remain different from Pavlodar microfinance, so it tests pipeline portability rather than local validity.

Model reporting follows the principle behind Model Cards: intended use, evaluation context, limitations, and ethical considerations should accompany performance results [5]. ROC analysis is used for ranking performance [6], while probability quality is also examined with Brier score and calibration tables because ranking alone does not establish reliable probabilities.

4. Data And Evidence Boundary

4.1 Experiment A: Synthetic Prototype Data

Experiment A uses `data/raw/credit_risk_dataset.csv`, containing 5,000 synthetic rows and 22 raw columns. The target `credit_risk` has 3,829 positive and 1,171 negative examples. Variables include income, balance, deposits, digital banking activity, debt, loan amount, open loans, and late-payment count.

The dataset is suitable for deterministic software tests, workflow design, and diagnostic experiments. It is not evidence about real borrowers. Its target and feature relationships may reflect assumptions embedded during synthetic data creation rather than relationships that would generalize to an MFI population.

The optional Pavlodar regional layer is also a scaffold. Administrative names and public regional context are separated from assumed fields such as digital access, distance, and seasonal-income risk. These assumptions are documented in `data/external/README.md` and must not be interpreted as measured borrower attributes.

4.2 Experiment B: Public UCI Benchmark

Experiment B uses the UCI Default of Credit Card Clients dataset, which contains 30,000 records from Taiwan and is distributed under CC BY 4.0 [3]. Senim Risk normalizes its fields into the same evaluation workflow but keeps all benchmark artifacts in a separate directory. Results are never merged with the synthetic experiment.

4.3 Leakage And Proxy Controls

The leakage-safe baseline removes identifiers and target-like or unrealistic fields:

- `customer_id`;
- `credit_score`;
- `loan_default_history`;
- `fraud_flag`.

`late_payment_count` is retained in the baseline so its influence can be measured rather than hidden. It is then removed in a dedicated thin-file stress test. Adjacent monetary, affordability, formal-credit, and digital-access signals are monitored as possible proxies, because they may represent wealth, infrastructure, or prior access rather than repayment reliability.

5. Experimental Method

5.1 Split And Preprocessing

Experiment A uses a stratified 80/20 train/test split with `random_state=42`. The held-out set therefore contains 1,000 records. Five-fold stratified cross-validation is computed on the training data. Numeric fields use median imputation and standardization; categorical fields use most-frequent imputation and one-hot encoding. All transformations are fitted inside scikit-learn pipelines to avoid training/test preprocessing leakage.

5.2 Models

Two baselines are evaluated:

- Logistic Regression with balanced class weights and up to 5,000 iterations;
- Random Forest with balanced class weights, 300 trees, and minimum leaf size

of 10.

Logistic Regression is the operational explanation baseline because its transformed feature contributions are additive and inspectable. Random Forest tests whether nonlinear relationships improve ranking. Neither model is treated as causal.

5.3 Metrics

ROC-AUC measures ranking across thresholds; 0.5 represents random ranking and 1.0 perfect ranking [6]. Accuracy, precision, recall, and F1 describe behavior at a chosen classification cutoff. Brier score measures squared probability error, with lower values preferred. Calibration tables compare mean predicted probability with observed outcome frequency inside ten bins.

No metric is sufficient by itself. In an imbalanced dataset, accuracy can remain high even when ranking is uninformative. A good ROC-AUC does not guarantee calibrated probabilities or an acceptable lending policy.

5.4 Ablation, Errors, And Segments

The ablation study retrains each model under six scenarios: a raw diagnostic ceiling, a leakage-safe baseline, removal of `late_payment_count`, behavioral features only, regional features only, and behavioral plus regional features. A Dummy Classifier provides a reference point.

Error analysis records false positives and false negatives at probability 0.50. Segment tables describe outcomes by gender, employment, district, and settlement type. Because these attributes are synthetic or simulated, the tables are diagnostic checks, not proof of fairness.

5.5 Decision Policies And Portfolio Uncertainty

A model probability is not a lending decision. Senim Risk defines a lower approve threshold, an upper decline threshold, and a manual-review interval between them. Four named policies expose different risk and inclusion postures.

Seeded Monte Carlo portfolio simulation then applies a selected policy to a scored portfolio under baseline, adverse, and severe scenarios. A shared macro shock creates correlated movement; application-level shocks represent residual calibration uncertainty; common random numbers support paired scenario comparison. The simulation does not alter borrower scores. Its monetary outputs remain prototype amount units, not calibrated KZT forecasts or regulatory VaR.

5.6 Model Robustness And Sampling Uncertainty

Model robustness is evaluated separately from portfolio simulation. First, the leakage-safe baseline and the no-`late_payment_count` ablation are retrained on ten deterministic stratified 80/20 splits. The resulting mean, standard deviation, and observed minimum-maximum describe sensitivity to split choice; they are not population confidence intervals.

Second, 1,000 stratified bootstrap replicates of the fixed held-out set produce 95% percentile intervals for ROC-AUC and Brier score. Third, models trained on the unmodified data score held-out records under three deterministic stresses: reduced digital activity, affordability pressure, and missing repayment history. The stress magnitudes are explicit synthetic perturbations, not economic forecasts. Metrics include probability movement and the share of classifications that cross the 0.50 diagnostic threshold.

6. Results

Research Finding 1: Moderate Synthetic Ranking

Model	Test ROC-AUC	Brier score	Precision	Recall	F1
Logistic Regression	0.806	0.186	0.896	0.712	0.793
Random Forest	0.830	0.143	0.986	0.641	0.777

Random Forest has the strongest held-out ranking and probability error, while Logistic Regression has higher recall and F1 at the default cutoff. The cross-validation ROC-AUC means are 0.828 and 0.822 respectively, showing that the single held-out ordering is not a universal model ranking.

Research Finding 2: One Feature Nearly Reproduces The Full Result

`late_payment_count` alone has directional ROC-AUC about 0.827, close to the full Random Forest's 0.830. It is also the largest Logistic Regression coefficient by absolute magnitude and accounts for approximately 65% of Random Forest impurity importance in the current artifact.

This does not prove that late-payment history is invalid. It shows that the synthetic result depends on a field that may be missing, sparse, or structurally different for the intended thin-file population.

Research Finding 3: Thin-File Ablation Collapses Ranking

Scenario	Logistic Regression ROC-AUC	Random Forest ROC-AUC
Raw diagnostic ceiling	0.966	1.000
Leakage-safe baseline	0.806	0.830
No `late_payment_count`	0.486	0.492
Behavioral only	0.499	0.494
Regional only	0.551	0.551
Behavioral + regional	0.547	0.529

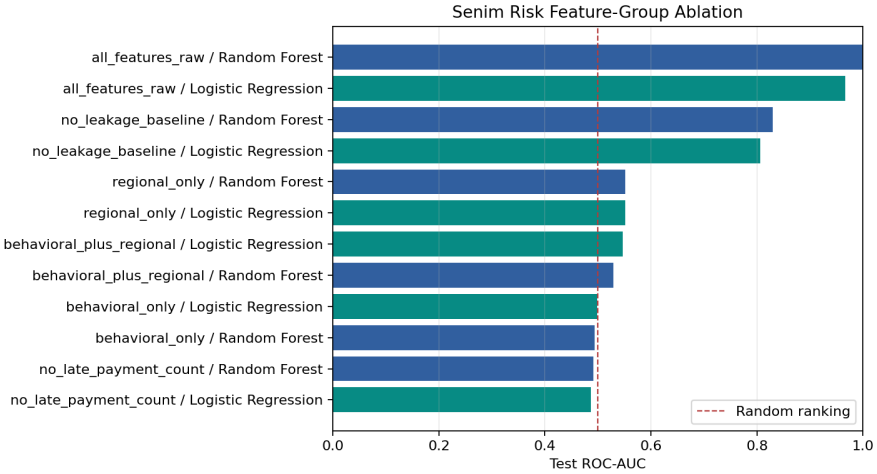


Figure 1. Held-out ROC-AUC by ablation scenario.

After `late_payment_count` is removed, both models rank near chance. Random Forest still reports accuracy 0.749 in that scenario, illustrating why accuracy alone is misleading under class imbalance: it can favor the majority class without learning a useful ranking.

The raw diagnostic ceiling also demonstrates why leakage checks matter. Near perfect performance disappears when target-like fields are excluded.

Research Finding 4: The Ablation Result Survives Resampling

Scenario	Model	Mean ROC-AUC	Standard deviation	Observed range
Leakage-safe baseline	Logistic Regression	0.826	0.011	0.806-0.844
Leakage-safe baseline	Random Forest	0.829	0.007	0.818-0.844
No late-payment count	Logistic Regression	0.469	0.017	0.444-0.492
No late-payment count	Random Forest	0.501	0.019	0.474-0.540

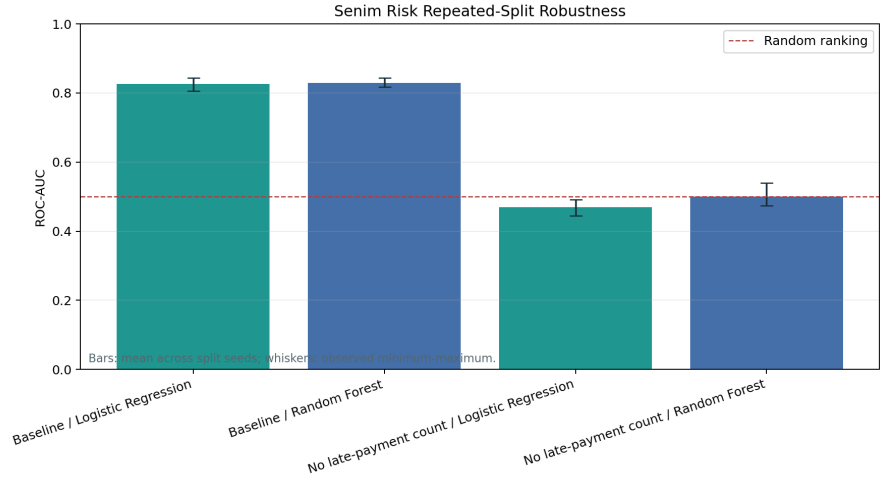


Figure 2. Mean held-out ROC-AUC and observed range across ten split seeds.

The fixed-split 95% bootstrap intervals tell the same story. Baseline ROC-AUC is 0.806 [0.781, 0.829] for Logistic Regression and 0.830 [0.804, 0.853] for Random Forest. Without `late_payment_count`, the intervals become 0.486 [0.445, 0.529] and 0.492 [0.448, 0.532], both spanning random ranking.

Input stress also reveals a deployment dependency. Removing repayment-history availability reduces ROC-AUC by 0.318-0.323 and moves the absolute predicted probability by about 0.284 on average. At the

diagnostic 0.50 threshold, 39.2% of Logistic Regression classifications and 50.2% of Random Forest classifications flip. The smaller digital-access and affordability stresses move at most 1.8% of classifications in this synthetic test. These results do not validate any stress magnitude; they show where data-availability monitoring must be concentrated.

Research Finding 5: The Public Benchmark Is Useful But Not Local Validation

Model	Test ROC-AUC	Brier score	Precision	Recall	F1
Logistic Regression	0.710	0.209	0.368	0.632	0.465
Random Forest	0.775	0.159	0.522	0.561	0.541

The UCI result shows that the same pipeline can process real public data and produce coherent model, calibration, feature, and error artifacts. It does not validate Pavlodar borrowers, microfinance products, local data collection, or local decision thresholds.

Research Finding 6: Error Costs Point In Different Directions

At threshold 0.50, the synthetic Logistic Regression produces 63 false positives and 221 false negatives on the 1,000-row test set. The false-positive rate is 0.269 and the false-negative rate is 0.289. In Senim Risk's risk label, a false positive may send a lower-risk borrower to unnecessary scrutiny, while a false negative may expose the lender to an unrecognized higher-risk case.

The most confident false negatives often have `late_payment_count = 0`, which is consistent with the ablation finding: the remaining synthetic variables do not provide enough independent signal.

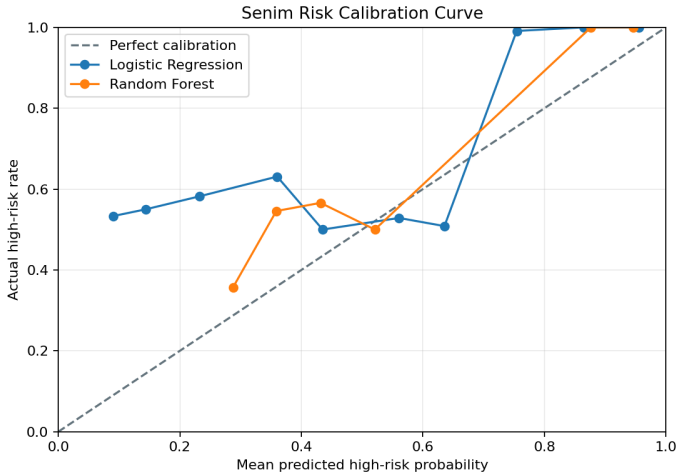


Figure 3. Calibration of held-out synthetic probabilities.

Research Finding 7: Policy Choice Changes The Product Outcome

Policy	Approve	Review	Decline	Share of all high-risk cases approved
Lender protective	12.7%	26.5%	60.8%	9.3%
Balanced review	27.8%	24.3%	47.9%	20.6%
Inclusion first	39.2%	27.0%	33.8%	28.9%
Starter-loan review	23.9%	39.4%	36.7%	17.2%

The table exposes a real systems trade-off. More permissive thresholds increase automatic access but also approve a larger share of high-risk cases. A wider review band reduces automatic decisions but requires analyst capacity. These figures are scenario outputs, not recommended production policies.

Research Finding 8: Unconstrained Profit Optimization Is Not A Sufficient Objective

Under the current illustrative interest margin and loss-given-default assumptions, the nominal profit-optimal threshold approves no applicants. This is a warning about both the dataset and the objective: optimizing only the modeled financial result can eliminate access. Senim Risk therefore reports minimum-approval constraints, segment outcomes, and review policies rather than presenting one profit-maximizing cutoff as the answer.

7. Product And Engineering Integration

The research result is connected to a working system rather than left in a notebook. The static frontend supports borrower, MFI analyst, and administrator workspaces. Public mode uses a browser-local synthetic API; local mode calls a FastAPI backend with typed schemas and SQLite persistence. The backend records application lifecycle, model version, score provenance, human decisions, audit events, and tenant scope.

The analyst view presents probability, risk band, local factors, warnings, scenario comparison, affordability indicators, and decision history. Model activation does not rewrite earlier decisions: older score packets are marked stale while retaining their original provenance. Monte Carlo runs store their seed, assumptions, portfolio fingerprint, model version, and result for later comparison.

This architecture does not make the model valid, but it makes the model's use inspectable. The public demo proves interface behavior only; it does not prove backend deployment, production security, or real-data readiness.

8. Threats To Validity And Ethical Boundary

The main limitations are substantive, not cosmetic:

1. Synthetic target dependence. Experiment A may reproduce assumptions from

data construction rather than borrower behavior.

2. No local outcome validation. No consented Kazakhstan MFI repayment data

has been used.

3. Benchmark mismatch. UCI records concern Taiwan credit-card customers,

not thin-file microfinance applicants.

4. No temporal test. The synthetic split is random, so it cannot establish

stability across economic cycles or data drift.

5. Exploratory segment analysis. Synthetic demographic and regional groups

cannot establish real fairness or equal access.

6. Proxy and privacy risk. Digital behavior, location, income, and prior

repayment may encode protected or structural disadvantage.

7. Uncalibrated economics. Interest margin, loss given default, operating

costs, shocks, and amount units are explicit scenario assumptions.

8. Prototype infrastructure. Local authentication, SQLite, and the partial

PostgreSQL path are not production controls.

Senim Risk must not be used to approve, decline, price, or rank real borrowers. A future pilot would require data minimization and consent, legal and privacy review, independent validation, temporal and segment testing,

probability calibration, human appeal and override procedures, monitoring, incident response, and accountable institutional ownership.

9. Reproducibility

The core study can be reproduced from the repository:

```
.venv\Scripts\python -m microscore --reports
.venv\Scripts\python -m microscore --benchmark uci-default
powershell -ExecutionPolicy Bypass -File scripts\check.ps1
.venv\Scripts\python scripts\build_research_paper.py
```

Experiment A artifacts are stored in `reports/research-artifacts/`; Experiment B artifacts are stored in `reports/benchmark-artifacts/uci-default-credit-card-clients/`. The artifact manifest records 5,000 rows, target name, test size, calibration bins, and `random_state=42`, together with the ten robustness split seeds, bootstrap iteration count, confidence level, and covariate-shift scenarios. Source code, tests, model card, data statement, Monte Carlo methodology, and the claim boundary in `docs/PILOT_EVIDENCE_CLAIMS.md` are versioned with the paper.

10. Future Validation Plan

The next scientifically meaningful step is not a more complex model. It is better evidence:

1. define the outcome, observation window, and operational decision with an MFI;
2. validate the minimum-data schema and consent process before collection;
3. obtain privacy-reviewed, de-identified pilot or partner data;
4. use temporal and out-of-institution evaluation where feasible;
5. repeat leakage, proxy, ablation, calibration, error, and segment analysis;
6. compare Logistic Regression with nonlinear models using stability,

interpretability, and operational cost as well as discrimination;

7. estimate thresholds and portfolio stresses from observed local outcomes;
8. establish monitoring, appeals, overrides, and retraining governance.

11. Conclusion

Senim Risk's strongest result is a limitation discovered through ablation. A synthetic Random Forest appears promising at ROC-AUC 0.830, yet its ranking falls to 0.492 when one repayment-history proxy is removed. The public UCI benchmark confirms that the pipeline can operate on real data, but does not transfer validity to Pavlodar microfinance. Repeated splits, bootstrap intervals, and missing-history stress confirm that the proxy-dependence finding is not an accident of one held-out sample.

The project therefore treats credit scoring as an evidence and decision-system problem rather than a leaderboard problem. A defensible prototype must connect model metrics to feature provenance, probability calibration, policy trade-offs, human review, auditability, and explicit non-use boundaries. That is the central engineering lesson of Senim Risk.

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